Maximus Watkins

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EDUCATION

DPhil Engineering Biology Sep. 2025 – Present
University of Oxford, EPSRC & BBSRC Engineering Biology CDT

- **Thesis:** *Developing tools and frameworks for responsible, biosecurity-aware AI-guided biological design*.
- **Supervisors:** Prof. Idris Kempf, Prof. Jean-Baptiste Lugagne, Dr. Oliver Crook.

MEng Mechanical Engineering Sep. 2019 – Jun. 2023
University of Bristol

- **Modules:** Engineering Mathematics, Computer-Based Modelling, Behaviour of Dynamic Systems.

FUNDING & AWARDS

- EPSRC & BBSRC UKRI Doctoral Scholarship (£66,000 + 4-year stipend) 2025–2029
- University of Bristol Engineering Education Research Group Funding (£1,000) 2023
- MBDA Summer Internship Funding (£9,000) 2021, 2022

RESEARCH EXPERIENCE

Foundation Model Cell Tracking May 2026 – Nov. 2026
Lugagne Group, University of Oxford

- Benchmarked *CellSAM2*, a SAM2-based foundation model, against specialised deep-learning cell trackers (EmbedTrack, Trackastra, DeLTA) on phase-contrast *E. coli* microscopy videos.
- Retrained all models on GPU and profiled inference cost: CellSAM2 scored highest on Cell-HOTA and division accuracy, but its runtime scaled with cell count.

Molecular Buffering in Two-Component Signalling Feb. 2026 – May 2026
Papachristodoulou Group, University of Oxford

- Built deterministic ODE models (MATLAB) of an engineered two-component signalling system with DNA-, RNA- and regulator-level buffering, with and without dichotomous feedback.
- Compared output attenuation across buffering strategies and assessed stability with parameter-free methods (structural influence matrices, piecewise Lyapunov functions).

Electroactive Collagen Wound Dressings Jan. 2026 – Feb. 2026
Nair Group, University of Oxford

- Team project (4 students) assessing electroactive collagen films as dressings for infected chronic wounds, using an *in vitro* infected-wound degradation model and fibroblast RNA-seq.
- Work included building a custom voltage-divider rig to test electrical stimulation of *E. coli* under neutral and alkaline conditions, and a commercial viability analysis.

Tutorial Repository Pedagogy Jun. 2023 – Aug. 2023
Rowlandson Group, University of Bristol

- Designed faculty tutorial resources informed by tutor and student surveys and interviews.

Embedded Inductive Sensor Design (MEng project) Jan. 2023 – May 2023
Croxford Group, University of Bristol

- Designed and built encapsulated passive inductive sensors for ultrasonic non-destructive testing of concrete, including magnetic field modelling and signal-processing scripts for stakeholders.

Swarm Robot Platform Fabrication (BEng project) Jan. 2022 – May 2022
Conn Group, University of Bristol

- Designed a low-cost, insect-inspired hexapedal swarm robot with computer-vision gait analysis.

INDUSTRY EXPERIENCE

Graduate Guidance, Control and Navigation (GCN) Engineer Aug. 2023 – Sep. 2025
MBDA UK

- **Collaborative Controller Development** *2025*
  - Aided in development of collaborative multi-aircraft control schema.
  - Produced simulations under various test harnesses and scenarios.
- **Object Clustering & State Estimation** *2024–25*
  - Clustered object behaviour data using RNN-based variational autoencoders.
  - Built novel state estimators fusing trial data and classifier outputs.
- **Hardware and Software Development** *2024*
  - Wrote backend code for control algorithms in Ada, bridging GCN and software.
  - Developed hardware-in-the-loop tests and documentation.
- **Control Robustness & Plume Effectiveness** *2023–24*
  - Verified control and aerodynamic algorithms via 6-DOF Monte Carlo simulation.
  - Built bespoke tools for processing wind-tunnel data.

GCN Intern Jun. 2022 – Sep. 2022
MBDA UK

- Designed a controller for a novel inertial test rig, auto-coded from Simulink to C++.

Image Processing Intern Jun. 2021 – Sep. 2021
MBDA UK

- Compared PointNet++ and PointGNN point-cloud segmentation (KITTI) with custom dataloaders.

CONFERENCE PRESENTATIONS

Synthetic Biology UK Nov. 2026
Southampton, UK

- **M. Watkins**, O. O'Connor, M. Dunlop, I. Kempf, J-B. Lugagne. *CellSAM2: Benchmarking a Foundation Tracking Model against Specialised Trackers on 2D* E. coli *Microscopy Data*. Poster.

Engineering Biology Summer School Jun. 2026
Oxford, UK

- **M. Watkins**, A. Papachristodoulou. *Structural analysis of molecular buffering and dichotomous feedback in an engineered two-component signalling model*. Poster.

Horizons in STEM Conference Jun. 2024
Bristol, UK

- J. Rowlandson, H. Lindskog, J. Bramton, R. Selwyn, S. Akromah, Y. Zanker, E. Brady, **M. Watkins**. *Innovating Personal Tutorials: Building an Activity Repository for Enhanced Student Engagement and Support*. Workshop.

Bristol Institute for Learning and Teaching (BILT) Conference Jul. 2023
Bristol, UK

- **M. Watkins**. *Tutorials Repository: Crafting reference resources for engineering tutorials*. Poster.
- E. Brady, **M. Watkins**, Y. Zanker, J. Ross, R. Selwyn. *Students as Partners in Engineering education research*. Presentation.

TEACHING & OUTREACH

Responsible Research and Innovation Facilitator Sep. 2026 – Present
University of Bristol

- Help to facilitate and plan bespoke workshops on RRI for CDT students.
- Involves holding discussions around research ethics, public engagement and critical thinking within academic research.

Open City Lab Public Engagement Project Apr. 2026 – Jan. 2028
We The Curious

- Designing a public engagement exhibition with the We The Curious science centre in Bristol, centred around care for Engineering Biology technologies.

Flying Start Challenge Sep. 2023 – Aug. 2025
STEM Ambassadors UK

- Teach aerodynamics to Year 7–9 pupils for regional glider design competitions; on committee, curate school content and help allocate a £100,000+ budget.

Peer-Assisted Study Session (PASS) Leader Sep. 2020 – Aug. 2021
University of Bristol

- Led flipped-classroom tutorials for first-year Mechanical Engineering undergraduates.

SKILLS

Programming	Python (PyTorch, NumPy, pandas), MATLAB & Simulink, Bash, LaTeX.
Methods	Deep learning & computer vision, ODE modelling, control design, Monte Carlo.